BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 143 · Lesson 143 · The TCP/IP Stack and Encapsulation

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which device typically forwards traffic between different IP networks?

Q2. Which logic gate outputs 1 only when both inputs are 1?

Q3. What is the accumulator for?

Q4. Which is faster in the worst case on sorted arrays, linear or binary search?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

application layer — Protocols providing services to programs, such as HTTP or DNS.

transport layer — End-to-end service such as TCP or UDP.

internet layer — IP packet addressing and routing between networks.

link layer — Local frame delivery on physical or wireless link.

New learning

Application data — HTTP sends a request body/application message.

Transport and IP — TCP segments carry ports, IP packets carry network addresses.

Frame on local link — Ethernet/Wi-Fi frame delivers packet over one link, then changes at next hop.

Worked example

A browser sends an HTTP request using TCP/IP over Ethernet. Describe encapsulation from top to bottom.

1. HTTP application data — Identify the relevant information and make a prediction.

2. TCP segment in IP packet — Explain the change using correct technical terms.

3. Ethernet frame — HTTP application bytes become TCP segment payload; the TCP segment goes into an IP packet; the packet is carried in an Ethernet frame. Receivers process headers in reverse to deliver data to the browser/server.

We Do · Practise with guidance

A packet passes through a router from Ethernet into a Wi-Fi network. Which information changes?

Incoming Ethernet frame

Identify the relevant information and make a prediction.

Route IP packet

Explain the change using correct technical terms.

Outgoing Wi-Fi frame

Router removes the incoming link-layer framing, processes/forwards the IP packet and builds new outgoing link-layer framing. IP hop limit/TTL normally decreases; end-to-end transport data need not change.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Name four layers of the TCP/IP stack and give one role for each. [8]

Question 2

Describe encapsulation of application data over TCP/IP and Ethernet. [6]

Question 3

Explain what a router changes when forwarding a packet to a different link technology. [8]

Exit Ticket · Answer before checking

Every router preserves the same link-layer frame all the way to the destination.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Router.

Q2. AND.

Q3. Holding a CPU's working data/ALU results.

Q4. Binary search: O(log n) rather than O(n).

You Do pages · Independent questions

Q1. Application user-facing network protocols [2], transport end-to-end delivery/ports [2], internet addressing/routing [2], link local frame transport [2].

Q2. Application message [1], TCP header/segment [2], IP header/packet [2], frame encapsulation [1].

Q3. Remove old link frame [2], inspect IP destination/routing [2], reduce TTL/hop limit and update necessary header fields [2], encapsulate with new link frame [2].

Exit Ticket · Model answer

False. Frames are recreated for each local link/hop.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: